# CSE 276A Homework 2: Closed-Loop Waypoint Driving with Object Landmarks

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**Video:** https://youtu.be/Xsz-RKL-paU?feature=shared

I took the extra credit option: instead of AprilTags, I trained a detector on my own household objects. The car drives $(0, 0, 0) \to (1, 0, 0) \to (1, 2, \pi) \to (0, 0, 0)$ using those objects for visual feedback.

## Coordinate frames and start pose

**World:** the origin is the middle of the car at the start, $+x$ is the car's initial heading, and $+y$ is to its left. **Vehicle:** the origin is the car's turning center, taken as the middle of the 25 cm body, with $+x$ forward and $+y$ left. **Sensor:** the camera sits at the very front, 12.5 cm ahead of the turning center, and points about 7° left of the car's axis. Detections are expressed as forward and left distances from the lens. **Start pose:** $(0, 0, 0)$.

## (a) Closed-loop controller

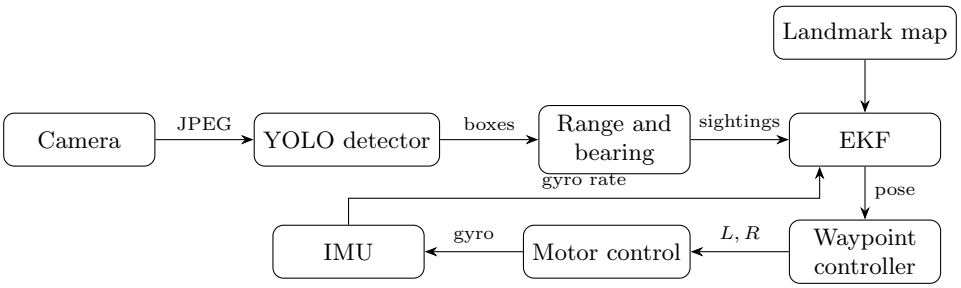



Figure 1: System graph. The commanded wheel values also feed the EKF prediction.

**Camera** publishes 1280×720 JPEG frames at about 10 Hz. **YOLO detector** finds landmarks and sends pixel boxes at 5 Hz. **Range and bearing** turns each box into a position relative to the camera. **Landmark map** holds the known landmark positions. **EKF** fuses gyro rate, commanded speed, and sightings into a pose. **Waypoint controller** turns the pose into wheel commands. **Motor control** sends them to the board over serial and publishes the IMU.

The controller keeps the turn, drive, turn structure from HW1, but the drive is now closed loop. It makes one continuous in-place turn if the goal is more than 15° off the heading, then steers toward the goal with a proportional heading term, capped at 0.1 of wheel command. Within 0.4 m of the goal it holds its heading, because the bearing to a nearby goal is too noisy to steer by. It stops when the EKF says it has passed the goal, or when a landmark straight past the goal is seen at the range it should have at the goal. After stopping it looks for 1.5 s to get clean sightings, and drives on if it is still short and the goal is within 60° of ahead.

## (b) Landmarks

Every leg has a landmark ahead of it, because the car only sees what is in front of it. Landmarks sit outside the path, since a landmark placed on a waypoint gets hit. Moving the car in practice showed this: it bumped the toilet paper at $x = 1.2$ and the cereal box when it was too close to the origin, so I moved both out. The cereal box is placed by its measured distance from the origin to its front face (30 cm), assuming it sits on the return line.

| Landmark | Position (m) | Why it is there |
|---|---|---|
| Toilet paper | $(1.24, 0)$, map $(1.35, 0)$ | Straight ahead on leg 1, and still in view at $(1, 0)$ |
| Dog plushie in a cup | $(1.2, 1)$ | Beside the middle of leg 2, seen from both ends |
| Water bottle | $(1.2, 2)$ | Ahead on leg 2, helps fix heading with the dog |
| Cereal box | $(-0.15, -0.30)$ | Straight past the origin on the way back |

# (c) Estimating pose and handling noise

The state is $(x, y, \theta)$. I used an EKF on a known map because it combines sensors that fail differently: dead reckoning is smooth but drifts, and vision does not drift but is noisy and sometimes missing. A single sighting cannot give a full pose anyway, so it needs a filter that keeps the rest.

**Measurement.** Lens distortion is removed from the box first. In normalized coordinates a vertical landmark of height $H$ spans $H/Z$, so

$$Z = \frac{H}{y_{\text{bottom}} - y_{\text{top}}}, \qquad X = Z\, x_{\text{center}}, \qquad r = \sqrt{X^2 + Z^2}, \qquad \beta = \operatorname{atan2}(-X, Z).$$

Intrinsics come from my chessboard calibration ($f_x = 648.25$, $f_y = 648.44$, $c_x = 639.82$, $c_y = 306.78$, RMS 0.717 px). Boxes touching the image border are dropped, because their height is not the landmark's height. The range is then corrected to the landmark center as seen from the car's turning center.

**Prediction.** With commanded speed $v$ and gyro rate $\omega$,

$$x_{t+1} = x_t + v\,\Delta t \cos\bar{\theta}, \quad y_{t+1} = y_t + v\,\Delta t \sin\bar{\theta}, \quad \theta_{t+1} = \theta_t + \omega\,\Delta t.$$

Process noise grows with distance driven and angle turned, not with time, because a stopped car does not move.

**Update.** Each sighting is a range and bearing to a landmark at a known position, fused with the usual EKF update. Measurement noise is 5% of range and 2° of bearing, with at least 8 cm of range noise while moving.

**Noise handling.** These all came from real failures:

- *Outliers:* a sighting that fails a chi-square gate (13.8 on 2 degrees of freedom) is rejected. Three in a row stop the run.
- *Image latency:* a frame is about 0.3 s old. Ignoring this made the car overshoot by 11 cm, so the distance covered in that time is subtracted while driving, and sightings are skipped during turns and for 0.5 s after any turn or stop.
- *Correlated errors:* frames a moment apart repeat the same box error, so each landmark counts at most every 0.5 s.
- *Frozen gyro:* the IMU sometimes repeats one value for seconds while driving. Integrating it made the EKF see a turn that was not happening, and the controller then steered the real car off course. After 5 identical samples the rate comes from the wheel commands instead, and heading noise is raised so sightings can correct it.
- *Calibration from data:* fitted effective landmark heights, a 3 cm range offset, the 7° camera yaw, a true speed of 0.31 m/s, and a wheel trim for the car's pull to the left.

**Safety:** the run stops itself on three rejections in a row, a failed turn, position uncertainty over 0.25 m, or 120 s.

# (d) When no landmark is visible

With nothing in view, the EKF keeps predicting from the gyro and commanded speed, and its covariance grows. The first good sighting afterward then gets a large gain and pulls the estimate back. The gate stays open because of that larger covariance. This matters at $(1, 2, \pi)$, where nothing is in view for the last part of leg 2 and for the turn. The heading uncertainty there reached 17.5° in the final run, then dropped once the cereal box came into view on the way back.

## (e) Detection method, robustness and accuracy

**Why this method.** I fine-tuned YOLO11n, a small one-stage detector, because the household objects needed a real detector, and because a boxed height is all I need for range. It takes a 416 pixel input and runs on the Rubik Pi CPU through ONNX Runtime, pinned to the fast cores. It takes about 45 ms per frame, and I cap it at 5 Hz to control heat, since the board reached about 80°C.

**Training.** I took 180 labeled images from the rover's camera at floor height, plus 25 empty-room images to teach it what is not a landmark. A first model trained on 60 images labeled the next batch for me to correct. Train and validation were split by consecutive blocks, not at random, since neighboring frames are nearly identical. I trained six classes; the mug and fan were not used in the final field.

**Accuracy.** I held out 80 new images in the real layout, labeled from scratch so the model's boxes could not influence the labels, and never trained on them: Most misses were boxes cut off by the image border, which I

| Class | Precision | Recall | mAP50 | mAP50-95 |
|---|---|---|---|---|
| Water bottle | 0.928 | 0.872 | 0.875 | 0.569 |
| Toilet paper | 0.726 | 0.706 | 0.777 | 0.394 |
| Cereal box | 1.000 | 0.924 | 0.935 | 0.611 |
| Dog plushie | 0.974 | 0.930 | 0.925 | 0.792 |
| All | 0.907 | 0.858 | 0.878 | 0.592 |

reject anyway. For uncut boxes, the bottle, box and dog were found every time, at sizes down to about 2 m for the bottle and box and about 1.4 m for the dog. Against a tape measure, range read 2 to 4 cm long from 0.3 to 2 m, which led to the 3 cm correction. Box height differed from my labels by 2.3% at the median and 9.6% at the 90th percentile.

**Robustness.** The weak spots are real. A white wastebasket was detected as toilet paper with 0.79 confidence, so I removed it from the field. The dog was missed at 2 m and small toilet paper at 41 px, so both are only reliable at shorter range, and the map keeps them close to the path. The test images come from the same room as the training images, so these scores do not promise anything about a new room or lighting.

## (f) Smoothness

The car drives at a steady 0.3 command and steers only gently. The first versions rocked left and right to fine-tune each turn, spun around to recover, and made a needless turn at the start. I removed all of that: each turn is now one continuous motion, small heading errors are steered out while driving, and it never spins to fix a small error. The cost is a heading error of several degrees left at each waypoint, because nothing corrects the end of a turn. The car also stops for about 1.5 s at waypoints to look, so it is smooth within a leg but not a continuous motion.

## (g) Total moving distance

5.39 m of driving, or 10.8 units with a 0.5 m unit, plus 317° of in-place rotation, in about 34 s of motion. Both are summed from the EKF pose log.

## (h) Error at each waypoint

The run used 63 sightings and rejected none. The EKF numbers are the car's own estimate, so the measured columns are what count. At $(1, 2)$ the car stopped about 16 cm from the water bottle, against 15 cm expected from the map.

| Waypoint | EKF position | EKF error | Measured position error | Measured heading error |
|---|---|---|---|---|
| $(1, 0, 0)$ | $(1.059,\ 0.002,\ +7.8^\circ)$ | 5.9 cm, 7.8° | About 10 cm short, see below | 5° |
| $(1, 2, \pi)$ | $(0.923,\ 2.001,\ 175.9^\circ)$ | 7.7 cm, 4.1° | About 1 cm, see below | 2° |
| $(0, 0, 0)$ | $(0.026,\ -0.026,\ -14.5^\circ)$ | 3.7 cm, 14.5° | About 0: stopped right on the mark | 10° |

**Human error at** $(1, 0)$**.** Before the final run I accidentally moved the toilet paper to $x = 1.24$, while the map still had it at $x = 1.35$, so the landmark table lists both. The car localizes against the map, so a landmark 11 cm closer than expected made it believe it had gone further than it had, and it stopped at about $x = 0.9$ instead of $x = 1.0$. This is a setup mistake, not a failure of the localization or the controller: the car did what the map told it to, and it recovered to finish on the origin once the other landmarks came into view.

## (i) Trajectory

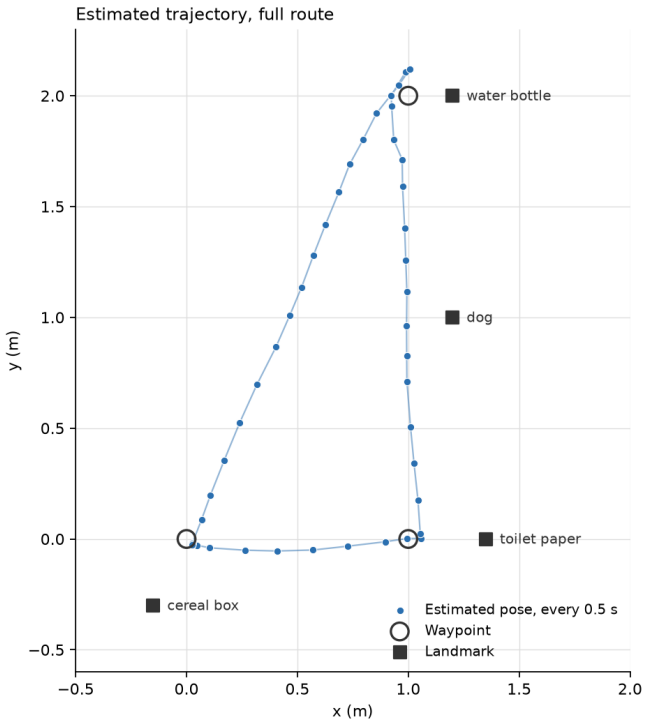



Figure 2: EKF position every 0.5 s in the final run. Near $(1, 2)$ the extra points past the waypoint are the estimate being corrected by the first cereal box sightings on the way back, not necessarily the car moving there.

## (j) Reflection

This worked much better than open loop, and it finished on the origin. The final run is the best of many attempts, so it does not show what every run would do. Most of the error I chased turned out not to be in the filter. The two biggest causes were a gyro that freezes while driving and landmark positions that did not match the map, and the second looked exactly like a control problem. When I stopped tuning and measured the real position of the cereal box, the car stopped ending short. The same thing caused the 10 cm shortfall at $(1, 0)$ in

the final run, when I bumped the toilet paper out of place. The whole method depends on the map being right, so I should recheck landmark positions right before each run. The next run after the official one also finished well by its own estimate, which suggests the setup is repeatable, though I did not measure that run the same way.

Several things are still weak. I never found out why the gyro freezes, so I only detect and work around it, using a wheel model that is rough. The car stays blind for the last part of leg 2 and the turn at $(1, 2)$, and another landmark there would help. Final heading is the weakest number, since the car takes one turn and does not correct it. With more time I would fix the IMU first, add a landmark near $(1, 2)$, and test in another room.